Calculated Fields, Parameters & LOD Expressions

Question 1 : What is a Calculated Field and why is it important in business dashboards?

- *Definition: A Calculated Field is a new field created by applying a formula or calculation to existing data. It helps users analyze data and generate additional insights without changing the original dataset.*

Example: In a sales dashboard, a calculated field can be used to calculate profit:

$$Profit = Sales - Cost$$

If Sales = ₹10,000 and Cost = ₹6,000, then Profit = ₹4,000.

Importance in business dashboards:

1. *Improves analysis: Helps calculate important metrics such as profit, profit margin, and growth rate.*
2. *Saves time: Automates calculations instead of calculating values manually.*
3. *Supports decision-making: Helps managers identify profitable products and business trends.*
4. *Provides better insights: Converts raw data into meaningful information for reports and charts.*
5. *Enables dynamic analysis: Calculated values can update when the underlying data changes.*

Question 2 : Why can't parameters change visuals directly in Tableau?

- *In Tableau, parameters are standalone values that allow users to input or select a value, such as a number, date, or category. However, parameters cannot change visuals directly because they are not automatically connected to the data or charts.*

To affect a visual, a parameter must be connected to a calculated field, filter, or reference line.

Example: A sales dashboard has a parameter that lets users select a sales target of ₹50,000 or ₹1,00,000. The parameter alone will not change the chart. A calculated field must use the selected target to compare actual sales against it.

Why is this important?

1. *Parameters provide user control over dashboard analysis.*
2. *Calculated fields connect parameter values to data.*
3. *This combination makes dashboards interactive and flexible.*

Question 3 : When should EXCLUDE LOD be used?

- *In Tableau, EXCLUDE LOD (Level of Detail) is used when we want to perform a calculation at a level that excludes one or more dimensions from the current view.*

It allows us to calculate values at a broader level of detail, even when the visualization displays more detailed information.

Example:

Suppose a sales dashboard shows sales by Region and Product. We want to calculate the total sales for each Region, ignoring the Product dimension.

Tableau formula:

{ EXCLUDE [Product] : SUM([Sales]) }

This formula calculates regional sales totals without considering Product, allowing us to compare each product's sales with its region's total.

When should EXCLUDE LOD be used?

1. *When calculating regional totals while displaying individual products.*
2. *When comparing detailed values with overall or group totals.*
3. *When removing a dimension's effect from a calculation without removing it from the visualization.*

Question 4 : Difference between FIXED and INCLUDE LOD expressions?

- *In Tableau, FIXED and INCLUDE LOD (Level of Detail) expressions are used to calculate values at a specific level of detail, but they work differently.*

Basis	***FIXED LOD***	***INCLUDE LOD***
Meaning	*Calculates values using specified dimensions, regardless of the dimensions in the view.*	*Adds specified dimensions to the dimensions already present in the view for calculation.*
Dependency	*Independent of the view's dimensions.*	*Depends on the dimensions in the view.*

Purpose	*Calculates values at a fixed level of detail.*	*Calculates values at a more detailed level.*
Example	*Total sales by Region, regardless of the view.*	*Average sales by Customer within each Region.*

- *Examples:*
- *1. FIXED LOD:*
- *{ FIXED [Region] : SUM([Sales]) }*
- *This calculates total sales for each region, regardless of which dimensions are displayed in the visualization.*
- *2. INCLUDE LOD:*
- *{ INCLUDE [Customer Name] : SUM([Sales]) }*
- *This calculates sales at the customer level in addition to the dimensions already present in the view.*